

Opposing CoC moment during contact



Null-space torques and projector

Null-space projector:

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ (EE) velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

Damping torque:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Slows joint motion in the null space.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

Conditioning torque:

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

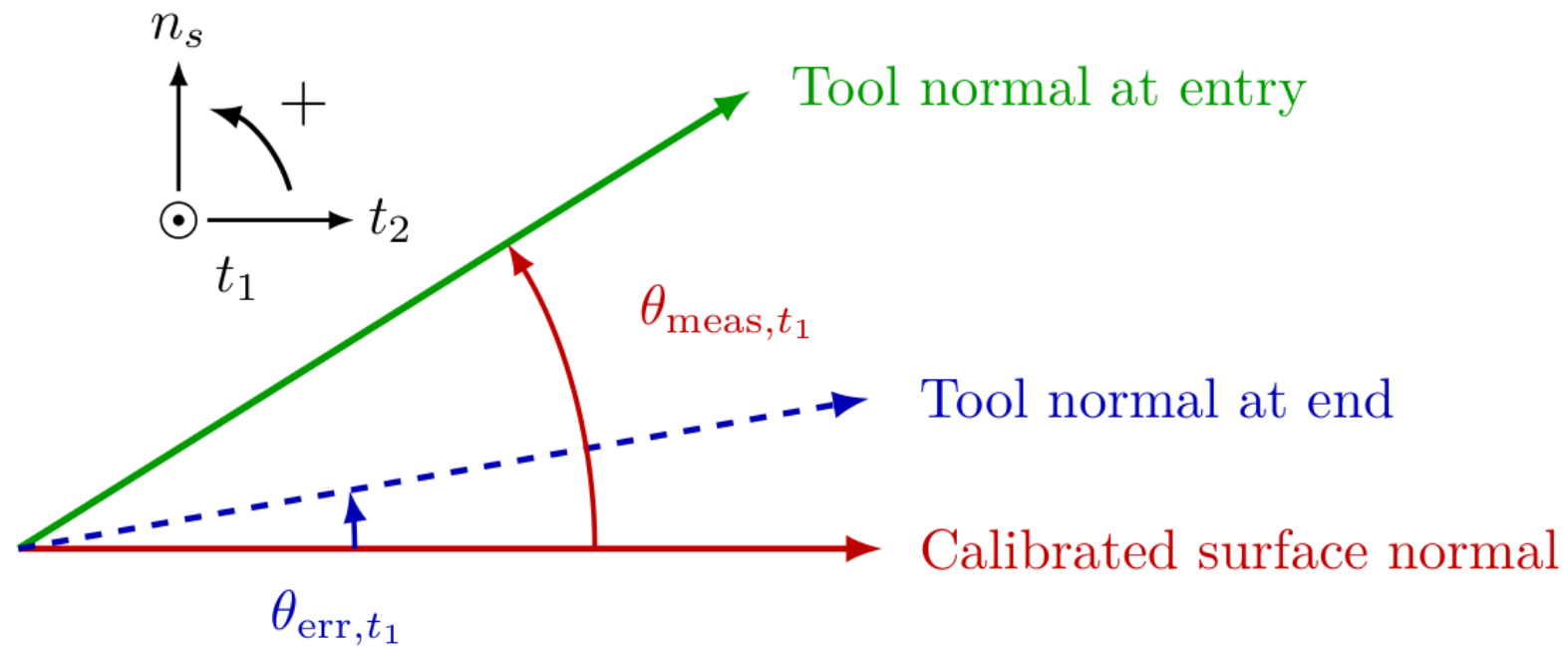
Seeks better local conditioning.
Can act at zero joint velocity.

k_σ : torque magnitude [N m]

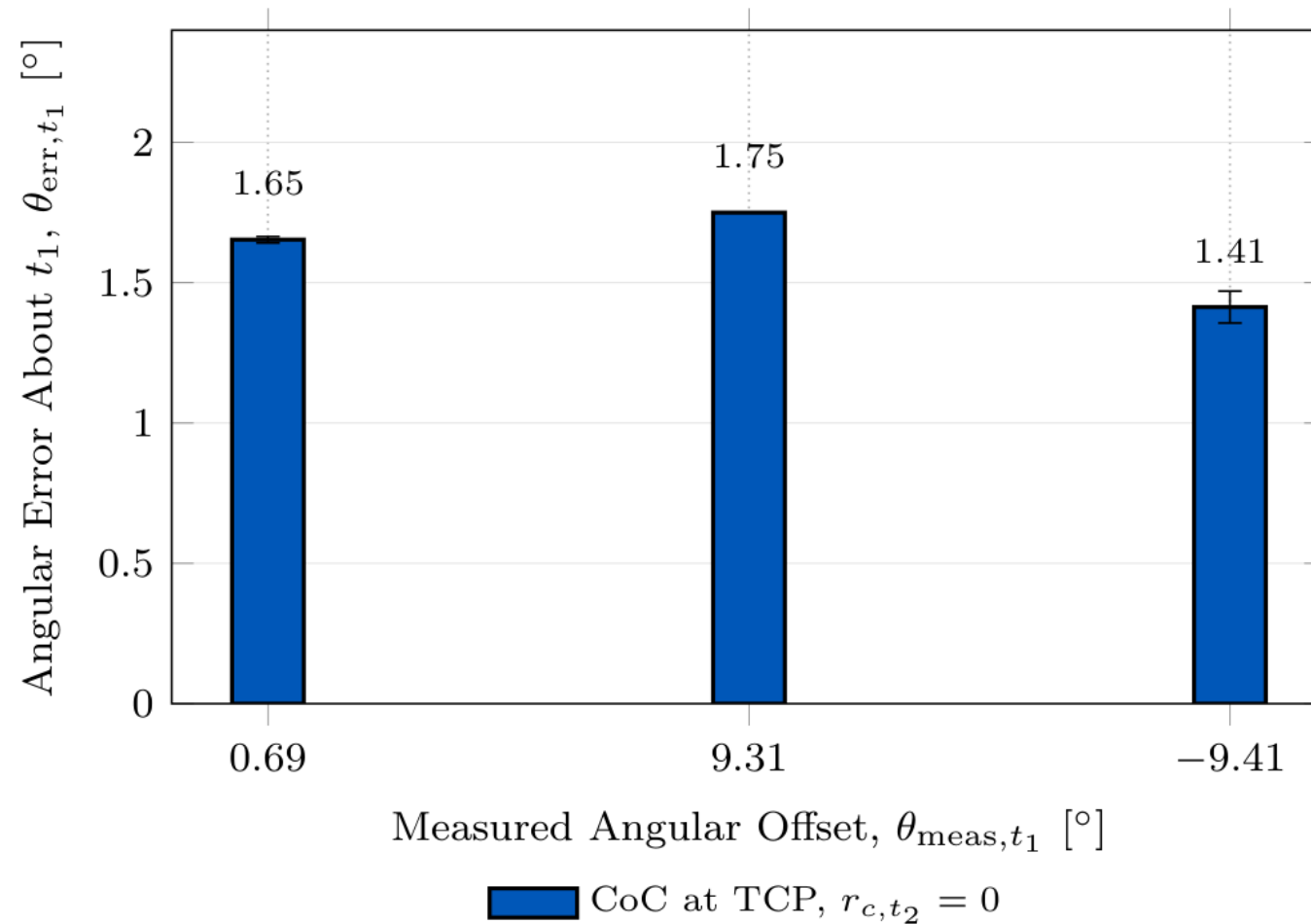
v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

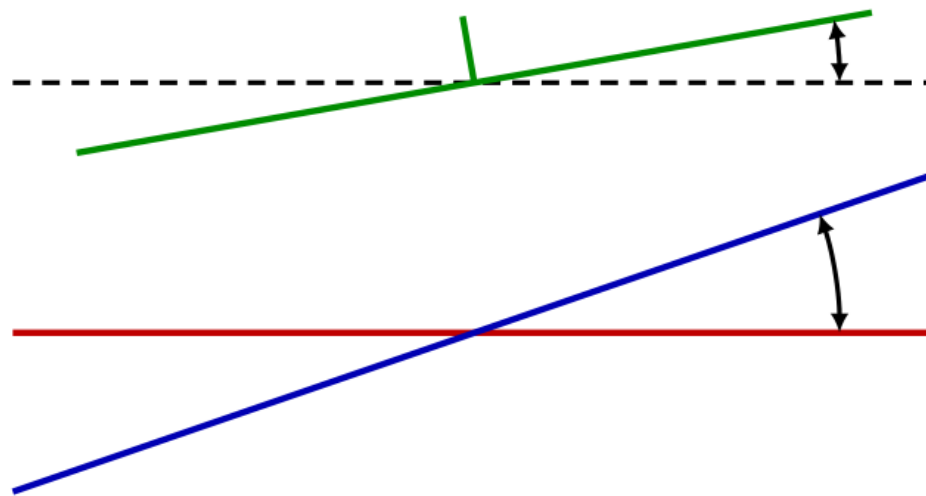
Entry offset and remaining angular error



Angular error with CoC at the TCP



Sources of angular offset



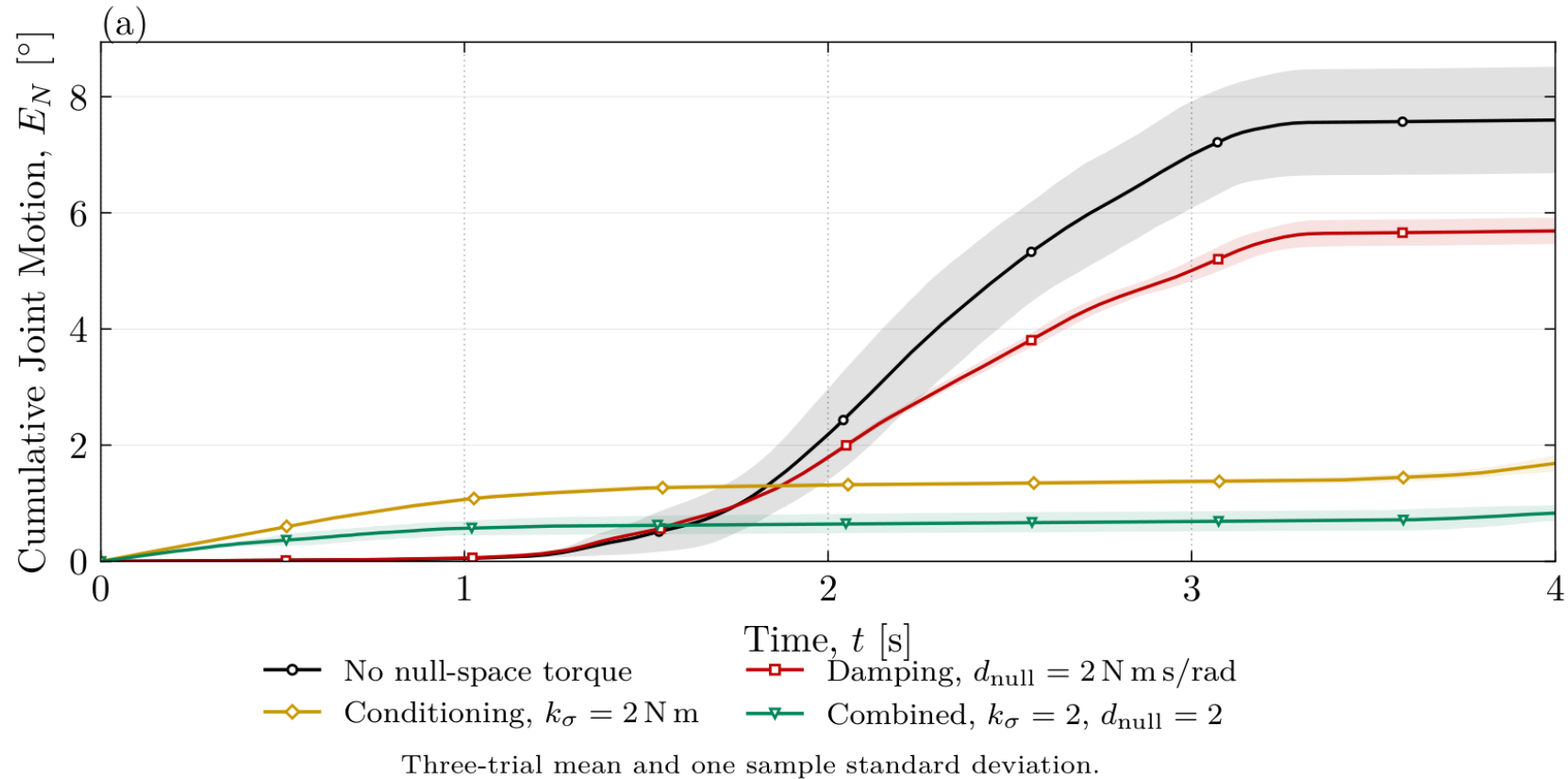
Desired–achieved difference

Configured–physical difference

— Configured surface
- - - - Desired orientation

— Physical surface
— Tool face

Damping reduces cumulative joint motion



≈25% less cumulative motion with damping alone

≈50% less cumulative motion with damping added to conditioning